INCH-POUND

MIL-C-20354C (AR)
AMENDMENT 1
8 March 1993

MILITARY SPECIFICATION

CARTRIDGE, 105MM, ILLUMINATING
M314A3, LOADING, ASSEMBLING AND PACKAGING

This amendment forms a part of MIL-C-20354C (AR), dated 19 May 1988, and is approved for use by the U.S. Army Armament, Munitions, and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 4

3.2.3, Burning time - candle power of the illuminant assembly (dwg. 9206749).: Change:

From:

"3.2.3 Burning time - candle power of the illuminant assembly (dwg. 9206749). The illuminant assembly during static testing shall not have a burning time of less than 77 seconds minimum (min.). The luminosity of all the illuminant assemblies shall not have an average candle power of less than 590,000 (min.).

To:

"3.2.3 Luminosity and burning time. The average luminosity of the canister assemblies shall be not less than 525,000 candlepower. The burning time of the canister assemblies shall be not less than 60 seconds. Determination shall be as specified in 4.4.3.5 and 4.5.3.

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PAGE 7

TABLE 1, First Article Inspection: Add the following new inspections to Iluminating Canister:

"Illuminant composition	test	4 ounces	3.2.1	4.5.2.2/ 4.5.2.3
Luminosity and burning	time	5 pair	3.2.3	4.4.3.5/ 4.5.3"

TABLE 1, First Article Inspection: Change the inspections of First Fire Composition:

From:

"First Fire Composition		4oz	3.2.1	4.4.3.4.1/4.5.2.1
Illuminating composition	test	4oz	3.2.2	4.4.3.4.1/4.5.2.2
Static functioning		5sets	3.2.3	4.4.3.5/4.5.3"

To:

"First Fire Composition				
First Fire Composition	test	4 ounces	3.2.1	4.5.2.1/4.5.2.3"

PAGE 16

4.2.2.7 Add the following Major Defect:

Major				
108	Plug disassembly torque, max. (if applicable)	0.40%	3.1	gage

PAGE 25

4.4.3.5 Static test of illuminant canister assembly
Change:

From:

"4.4.3.5 Static test of illuminant canister assembly. The following sampling plan will apply with the sample tested as specified in 4.5.3 and in accordance with Dwg. 9206749."

To:

"4.4.3.5 Luminosity and burning time (see 3.2.3 and drawing 9206749). The test shall be performed as specified in 4.5.3, and in accordance with Dwg. 9206749. A sample of two canister assemblies shall be selected in accordance with the sampling

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plan shown in 4.4.3.5. One canister assembly shall be tested for static luminosity and burning time. If the luminosity is less than 525,000 candlepower, the second candle shall also be tested. If the resulting average of the combined sample is less than 525,000 candlepower (luminosity), inspection lots represented by the samples shall be rejected (see 4.4.3.5.1.7). No one sample can exhibit a burning time of less than 60 seconds. The candlepower integration time shall be from the best continuous 55 seconds of burn time."

PAGE 26

4.4.3.5.1: Change:

From:

"4.4.3.5.1 Each inspetion lot will cocnsist of eighty five (85) candles produced consecutively by the mixing pressing operation and will be designated as one mix/batch (see 6.8) for identification purposes."

To:

"4.4.3.5.1 Sampling plan for luminosity and burning time. An inspection lot for the canister assembly shall be eighty-five candles produced consecutively from one mixing and pressing operation, and will be designated as one mix/batch for identification purposes (see 6.8). Smaller lot sizes will be accepted when dictated by interruption or shutdown of operations. The illuminant, as it is mixed, shall be dispensed into containers which shall be numbered and used in order."

4.4.3.5.1.1: Change:

From:

"4.4.3.5.1.1 Select a sample of five (5) candles from each inspection lot (85 candles). The canister assembly shall be observed for compliance with the requirements specified in 3.2.3. Continue this sampling frequency until 15 consecutive samples (15 groups of 5 samples) comply with the specified requirements."

To:

"4.4.3.5.1.1 Level one sampling. A sample of two candles from each inspection lot shall be selected at random. The canister assembly shall be tested for compliance with the requirements specified in paragraph 3.2.3. This sampling frequency shall be continued until fifteen consecutive samples of two candles comply with the requirements of paragraph 3.2.3."

4.4.3.5.1.2 Change:

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From:

"4.4.3.5.1.2 After successful completion of 4.4.3.5.1.1, reduce the sampling frequency to five candles from each group of six inspection lots (510 candles). The canister assembly shall be observed for compliance with the requirements specified in 3.2.3. Continue this sampling frequency until 15 consecutive samples (15 groups of 5 samples comply with the specified requirements. In the event that three consecutive lots fail to comply with the requirements, the sampling shall revert to 4.4.3.5.1.1."

To:

"4.4.3.5.1.2 Level two sampling. After successful completion of level one sampling, the sampling frequency may be reduced to two samples from each group of six inspection lots. The canister assemblies shall be tested for compliance with the requirements specified in paragraph 3.2.3. This sampling frequency shall be continued until fifteen consecutive samples of two candles comply with the requirements of paragraph 3.2.3. In the event that one lot fails to comply with the requirements, the sampling shall revert to level one sampling."

4.4.3.5.1.3: Change:

From:

"4.4.3.5.1.3 After successful completion of 4.4.3.5.1.2, reduce the sampling frequency to five candles from each group of fiteen inspection lots (1275 candles). The canister assembly shall be observed for compliance with the requirements specified in 3.2.3. Continue this sampling frequency until 15 consecutive samples (15 groups of 5 samples) comply with the specified requirements. In the event that three consecutive lots fail to comply with the requirements, the sampling shall revert to 4.4.3.5.1.2."

To:

"4.5.3.5.1.3 Level three sampling. After successful completion of level two sampling, the sampling frequenecy may be reduced to two samples from each group of fifteen inspection lots. The canister assemblies shall be tested for compliance with the requirements specified in paragraph 3.2.3. This sampling frequency shall be continued until fifteen consecutive samples of two candles comply with the requirements of paragraph 3.2.3. In the event that one lot fails to comply with the requirements, the sampling shall revert to level two sampling."

4.4.3.5.1.4: Change:

From:

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4.4.3.5.1.4 After successful completion of 4.4.3.5.1.3, reduce the sampling frequency to five candles from each eight hours of production or 1615 candles, whichever is less. The canister assembly shall be observed for compliance with the requirements specified in 3.2.3. In the event that three consecutive lots fail to comply with the requirements, the sampling shall revert to 4.4.3.5.1.3."

To:

"4.4.3.5.1.4 Level four sampling. After successful completion of level three sampling, the sampling frequency may be reduced to two samples from each shift (not to exceed ten hours) of production. The canister assemblies shall be tested for compliance with the requirements specified in paragraph 3.2.3. In the event that one lot fails to comply with the requirements, the sampling shall revert to level three sampling."

4.4.3.5.1.5: Change:

From:

"4.4.3.5.1.5 In the event of any production stoppage greater than one week, the sampling plan shall revert to 4.4.3.5.1.2 and in the event of production stoppage greater than two weeks, the sampling plan shall revert to 4.4.3.5.1.1. If the continuous mixing operation has not been shut down, but merely diverted to another item, the sampling plan shall revert to 4.4.3.5.1.2."

To:

"4.4.3.5.1.5. In the event of any production stoppape of less than two weeks, the sampling plan shall revert to level one, with only seven consecutive samples required to advance to each reduced level until the level of sampling in use when production is reached."

4.4.3.5.1.6: Change:

From:

"4.4.3.5.1.6 In the event of failure of any sample to comply with the requirements, each individual inspection lot within the mix/batch must be tested. Failure of an inspection lot to comply with the requirements shall result in rejection of that particular inspection lot."

To:

"4.4.3.5.1.6. If the mixing operation has not been shut down, but merely diverted to another item, and has not

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experienced any cause for rejection per the sampling plan of that item(s), the sampling level shall remain unchanged, and fifteen consecutive successful samples of two candles shall be required to advance to the next level, if applicable."

4.4.3.5.1. Sampling plan for luminosity and burning time: Add new subparagraph 4.4.3.5.1.7:

4.4.3.5.1.7 In the event of failure of any sample to comply with the requirements, each individual inspection lot within the mix/batch must be tested. Failure of an inspection lot to comply with the requirements shall result in rejection of that particular inspection lot."

PAGE 27

Paragraph 4.5.2.1. First Fire Composition (moisture content)

Change SPECIAL SOLVENT

From:

"Equal parts by volume of carbon tetrachloride and anhydrous methanol."

To:

"Equal parts by volume of dimethylformamide and anhydrous methanol."

PAGE 28

4.5.3, Static test of illuminant canister assembly: Change:

From:

"4.5.3 Static test of illuminant canister assembly. The discs shall be removed from the candle prior to testing. In calculating the candle power of any one illuminant assembly, the best continuous 60 seconds of illuminometer readings are to be used."

To:

"4.5.3 Luminosity and burning time. The discs shall be removed from the candle prior to testing. In calculating the candle power of any one illuminant assembly, the best continuous 55 seconds of illuminometer readings are to be used."

PAGE 29

Add the following paragraph:

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"5.3.1 Marking of projectiles packed without projectile fuzes. Packaging of projectiles will proceed in accordance with the applicable drawing. In cases where projectiles are packed separate from their projectile fuzes, each projectile's container and packaging box shall be marked in such a way so that it is clearly denoted on the outside of the container and packaging box, that the contents do not include a projectile fuze. Direction on how to mark containers and packaging boxes is explained on the applicable packaging drawing."

Custodian:
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